Lecture 30

Section 6.1

Inner Product

Definition: Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. The inner product of $\mathbf{u}$ and $\mathbf{v}$ is defined to be $\mathbf{u} \bullet \mathbf{v} = \mathbf{u}^T\mathbf{v}$. The result is a 1×1 matrix, and so we associate the inner product to a number in $\mathbb{R}$. More explicitly, if

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \qquad \text{and} \qquad \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix},$$

then the inner product of $\mathbf{u}$ and $\mathbf{v}$ is given by

$$\begin{pmatrix} u_1 & u_2 & \cdots & u_n \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \sum_{i=1}^{n} u_i v_i.$$

Remark: This inner product is referred to as a dot product. We will generalize the inner product in Section 6.7.

Theorem: Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and c a scalar. Then, the following properties hold.

1. $\mathbf{u} \bullet \mathbf{v} = \mathbf{v} \bullet \mathbf{u}$
2. $(\mathbf{u} + \mathbf{v}) \bullet \mathbf{w} = \mathbf{u} \bullet \mathbf{w} + \mathbf{v} \bullet \mathbf{w}$
3. $(c\mathbf{u}) \bullet \mathbf{v} = c(\mathbf{u} \bullet \mathbf{v}) = \mathbf{u} \bullet (c\mathbf{v})$
4. $\mathbf{u} \bullet \mathbf{u} \geq 0$, and $\mathbf{u} \bullet \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

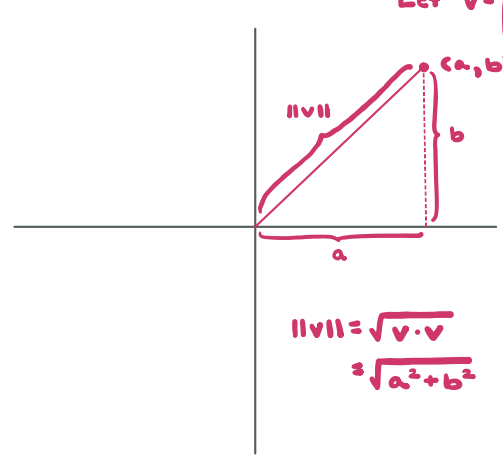
Length

Definition: Let $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in \mathbb{R}^n$. The length or norm of $\mathbf{v}$ is defined to be the nonnegative scalar

$$||\mathbf{v}|| = \sqrt{\mathbf{v} \bullet \mathbf{v}} = \sqrt{v_1^2 + \cdots + v_n^2},$$

where $||\mathbf{v}||^2 = \mathbf{v} \bullet \mathbf{v}$.

Let $v = \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$.



$||v|| = \sqrt{v \cdot v}$
$= \sqrt{a^2 + b^2}$

Remark: From the Pythagoreon theorem, it follows that $||\mathbf{v}||$ is the length of $\mathbf{v} \in \mathbb{R}^n$.

Useful Properties: Let $\mathbf{v}, \mathbf{u} \in \mathbb{R}^n$ and c a scalar.

1. $||c\mathbf{v}|| = |c|\,||\mathbf{v}||$ (*Proof:* We see that $||c\mathbf{v}||^2 = (c\mathbf{v}) \bullet (c\mathbf{v}) = c^2(\mathbf{v} \bullet \mathbf{v}) = c^2||\mathbf{v}||^2$. Taking square roots on both sides, we get the desired result.)

2. $||\mathbf{u} + \mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$

Definition: A vector whose length is 1 is called a unit vector. Let $\mathbf{v} \in \mathbb{R}^n$ be a nonzero vector. If we divide $\mathbf{v}$ by its length, i.e. we compute $\frac{\mathbf{v}}{||\mathbf{v}||}$, we obtain a unit vector. This process is called the normalization of $\mathbf{v}$. Note that $\frac{\mathbf{v}}{||\mathbf{v}||}$ is in the same direction as $\mathbf{v}$.

Claim: $\frac{\mathbf{v}}{||\mathbf{v}||}$ has length 1.

Proof: Let $\mathbf{v} \in \mathbb{R}^n$ and $c = ||\mathbf{v}||$. Notice that

$$\left|\left|\frac{\mathbf{v}}{c}\right|\right| = \sqrt{\frac{\mathbf{v}}{c} \bullet \frac{\mathbf{v}}{c}} = \sqrt{\frac{1}{c^2}(\mathbf{v} \bullet \mathbf{v})} = \frac{1}{\sqrt{c^2}}\sqrt{\mathbf{v} \bullet \mathbf{v}} = \frac{1}{||\mathbf{v}||}||\mathbf{v}|| = 1.$$

Example: Normalize $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ -2 \\ 3 \end{pmatrix}$.

Solution: We see that

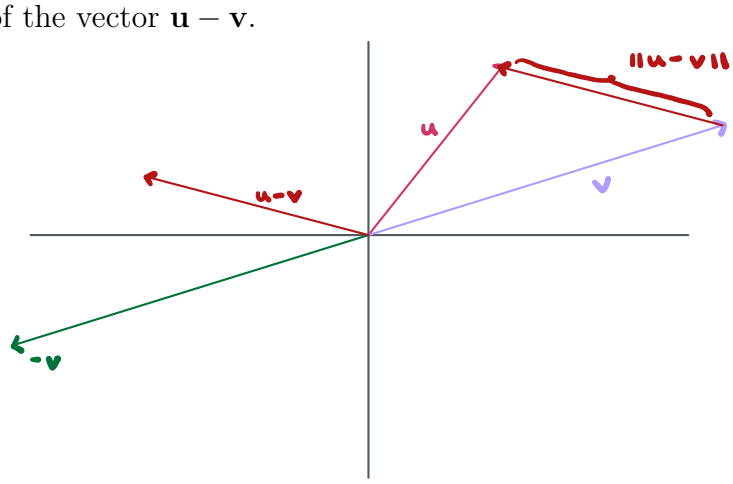
$$||\mathbf{v}|| = \sqrt{\mathbf{v} \bullet \mathbf{v}} = \sqrt{1^2 + 4^2 + (-2)^2 + 3^2} = \sqrt{1 + 16 + 4 + 9} = \sqrt{30}.$$

So,

$$\frac{\mathbf{v}}{||\mathbf{v}||} = \begin{pmatrix} 1/\sqrt{30} \\ 4/\sqrt{30} \\ -2/\sqrt{30} \\ 3/\sqrt{30} \end{pmatrix}.$$

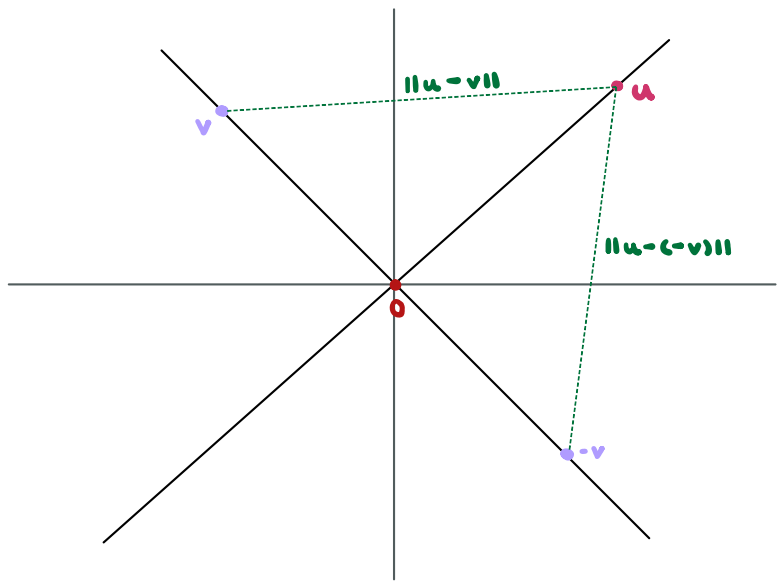
Distance in $\mathbb{R}^n$

Definition: Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. The distance between $\mathbf{u}$ and $\mathbf{v}$, denoted by $\text{dist}(\mathbf{u}, \mathbf{v})$, is $||\mathbf{u} - \mathbf{v}||$, i.e. the length of the vector $\mathbf{u} - \mathbf{v}$.



Orthogonality

Observation: Consider $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$. Draw the two lines through the origin determined by $\mathbf{u}$ and $\mathbf{v}$.



Note that these two lines are perpendicular if and only if $||\mathbf{u} - \mathbf{v}|| = ||\mathbf{u} - (-\mathbf{v})||$ if and only if $||\mathbf{u} - \mathbf{v}||^2 = ||\mathbf{u} - (-\mathbf{v})||^2$. We see that

$$\begin{aligned}
||\mathbf{u} - (-\mathbf{v})||^2 &= ||\mathbf{u} + \mathbf{v}||^2 \\
&= (\mathbf{u} + \mathbf{v}) \bullet (\mathbf{u} + \mathbf{v}) \\
&= \mathbf{u} \bullet (\mathbf{u} + \mathbf{v}) + \mathbf{v} \bullet (\mathbf{u} + \mathbf{v}) \\
&= \mathbf{u} \bullet \mathbf{u} + \mathbf{u} \bullet \mathbf{v} + \mathbf{v} \bullet \mathbf{u} + \mathbf{v} \bullet \mathbf{v} \\
&= ||\mathbf{u}||^2 + 2(\mathbf{u} \bullet \mathbf{v}) + ||\mathbf{v}||^2
\end{aligned}$$

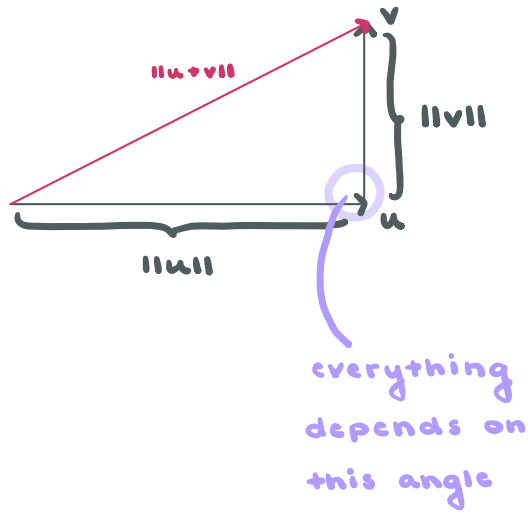
and

$$\begin{aligned}
||\mathbf{u} - \mathbf{v}||^2 &= (\mathbf{u} - \mathbf{v}) \bullet (\mathbf{u} - \mathbf{v}) \\
&= (\mathbf{u} + (-\mathbf{v})) \bullet (\mathbf{u} + (-\mathbf{v})) \\
&= \mathbf{u} \bullet (\mathbf{u} + (-\mathbf{v})) + (-\mathbf{v}) \bullet (\mathbf{u} + (-\mathbf{v})) \\
&= \mathbf{u} \bullet \mathbf{u} + \mathbf{u} \bullet (-\mathbf{v}) + (-\mathbf{v}) \bullet \mathbf{u} + (-\mathbf{v}) \bullet (-\mathbf{v}) \\
&= ||\mathbf{u}||^2 - 2(\mathbf{u} \bullet \mathbf{v}) + ||-\mathbf{v}||^2.
\end{aligned}$$

So, $||\mathbf{u} - \mathbf{v}||^2 = ||\mathbf{u} - (-\mathbf{v})||^2$ if and only if $2(\mathbf{u} \bullet \mathbf{v}) = -2(\mathbf{u} \bullet \mathbf{v})$ if and only if $\mathbf{u} \bullet \mathbf{v} = 0$. We have shown that two lines in $\mathbb{R}^2$ are perpendicular if and only if their defining vectors have a dot product of 0.

Let's generalize this idea to $\mathbb{R}^n$.

Definition: Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$. We say that $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $\mathbf{u} \bullet \mathbf{v} = 0$.

Note: The zero vector is orthogonal to every vector.

The Pythagorean Theorem: The vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if $||\mathbf{u}+\mathbf{v}||^2 = ||\mathbf{u}||^2 + ||\mathbf{v}||^2$.